

Combining strengths of Rubin, Euclid, and Roman (and radio surveys) for time-domain science and cosmology Notes

LST: observing in optical band

Roman: wide imaged

Problem: better depth $\rightarrow$ larger background star density

Modeling Rubin + Euclid together

- Covers wide range of bands

Transients: find them earlier

- Wide bands, wide parameters, lots of info about same object

Euclid $\rightarrow$ better Rubin photometry (less better shape)

DDP3: combines int of surveys

Space observations:

- Complex shape parameters
- Need more parameters
- Priors trained on machine learning

When degenerate:

- Model allows for finding more realistic cases

Scarlet 2:

- Allows for various kinds of analysis

DDP3 Pipeline

- Rather than correlating noise; looks at whole signal
- Combines data $\rightarrow$ better visuals

Can include grism spectra

- $\rightarrow$ How though? Hard to identify where emissions are coming from...
- May be too uncertain...

L55T:

- Will discover number of gradually: local supernovae (GLSN)
- Can extend these delays with Rubin data
 - $\rightarrow$ Roman should only get light curve, not time

Can add peer data in; mix with current literature/images

AGN not in galactic center:

- Candidates found via jitting light
 - $\rightarrow$ Not really that many
- TDEs used instead
 - Could look for AGNs that aren't accreting
- May also see via radio jets

Chargy-look AGN:

- Well documented can look at what's causing change
- Could be disk tearing up
 - $\rightarrow$ Need more verification
 - $\rightarrow$ Need to bridge from double peaked AGN
- Better statistics available $\rightarrow$ figure out what's normal
- Some have new radio jets $\rightarrow$ transient; not new long period

Questions:

- How will frequency of blackholes found in Gaia Data Release 3 change?
- What kind of data will be the most useful to add to this combined data analysis?

pipeline process